Ezra Song

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Education

University of Waterloo — Waterloo, ON
Bachelor of Mathematics (BMath), Honours Mathematics — *Sep 2025 – Apr 2029*

Technical Skills

Languages: TypeScript, JavaScript, Python, Swift, Java, C++, C#, SQL, Dart, HTML/CSS
Frameworks: React, Next.js, React Native, Expo, SwiftUI, Flutter, Node.js, NestJS, Fastify, PyTorch
Tools: PostgreSQL, Supabase, Docker, Git, GitHub Actions, Playwright, Vitest, REST APIs, OpenAPI, Linux

Professional Experience

Software Engineer Intern
Unmodal Research Inc. — *Jan 2026 – Sep 2026*

- Shipped responsive production features with Next.js, React, TypeScript, NestJS, and PostgreSQL, reducing UI duplication across frontend and backend systems.
- Worked on project orchestration layers; evaluated AI tools, frameworks, and competing workflows.
- Tested and deployed releases with Supabase, Docker, GitHub Actions, Vitest, and Playwright to catch regressions.
- Built iOS/macOS features with Swift and SwiftUI, integrating backend services for reliable workflows.

Web Developer
DoBetter.love — *Sep 2025 – Nov 2025*

- Built reusable React/JavaScript components and responsive HTML/CSS layouts from design feedback.
- Tested navigation, forms, error states, and breakpoints with browser tools; documented defect steps, environments, and severity for clearer handoffs.

Mobile App Developer
AIXFF — *Jul 2025 – Aug 2025*

- Built reusable screens, navigation, and responsive states in React Native/Expo for faster mobile prototyping.
- Modularized REST API calls and centralized state with React Context; tested user flows and device sizes to resolve edge cases and keep screen data consistent.

Junior Software Engineer
North P&D — *Feb 2023 – Nov 2023*

- Built reusable React/JavaScript and responsive HTML/CSS components for accessibility and consistent layouts.
- Fixed layout regressions with browser developer tools, cross-browser QA, and targeted CSS changes before release.

Projects

Sonolyth — *Flutter, Dart, Riverpod*

- Maintained an Android Spotube/Flutter fork: native playback, music plugins, synced lyrics, and Spotify metadata fixes.
- Versioned the app and HetuScript plugin in a monorepo; added nested playlist/album/artist folders, HTTP 429 retries, and preserved upstream sync.

Discord Automation Bot — *Node.js, discord.js, Docker*

- Built slash commands for moderation, reaction roles, internship alerts, and AMP server monitoring/control with instance autocomplete and permission checks.
- Added queued Lavalink music over WebSockets; deployed with Docker Compose/Unraid and bounded caches/V8 heap to keep idle memory near 30–40 MB.

Hackathons & Awards

UltraHacks 2026 — *University of Toronto* — Winner – Best Use of ElevenLabs (MLH)
UTMIST AI Tournament 2025 — *University of Toronto* — Top 8 Finalist
Go On Hacks 2025 — *University of Waterloo* — Best Pitch Award